ANALYTIC GEOMETRY

Polar Coordinates, Distance Formula, Area of Triangle & Conic Sections

Course Code: MATMJ52 — **Platform:** www.sciqb.com

1 Introduction to Polar Coordinates

In two-dimensional geometry, a point in a plane can be uniquely represented using a distance and an angle relative to a fixed reference point and direction. This system is known as the **Polar Coordinate System**.

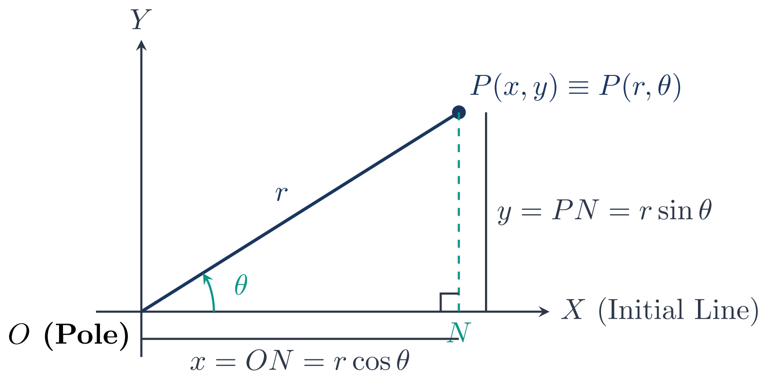
Definitions: Components of Polar Coordinates

- **Pole (or Origin)** $[O]$**:** A fixed reference point in the plane, denoted by O.
- **Initial Line** $[OX]$**:** A fixed horizontal straight line emanating from the pole O extending to the right.
- **Radius Vector** $[r]$**:** The line segment OP connecting the pole O to any point P in the plane. Its length is denoted by $r = |OP| \geq 0$.
- **Vectorial Angle** $[\theta]$**:** The angle measured counter-clockwise from the initial line OX to the radius vector OP. It is denoted by $\theta = \angle XOP$.

Thus, any point P is represented by the ordered pair (r, θ).

1.1 Transformation between Cartesian and Polar Coordinates

Let $P(x, y)$ be a point in the Cartesian coordinate system with the origin coinciding with the pole O and the positive x-axis coinciding with the initial line OX. Drop a perpendicular PN from P onto OX.



From the right-angled triangle $\triangle ONP$:

1. **Cartesian in terms of Polar Coordinates:**

$$ON = x = r\cos\theta \quad \text{and} \quad PN = y = r\sin\theta \tag{1}$$

2. **Polar in terms of Cartesian Coordinates:** Squaring and adding equations (??):

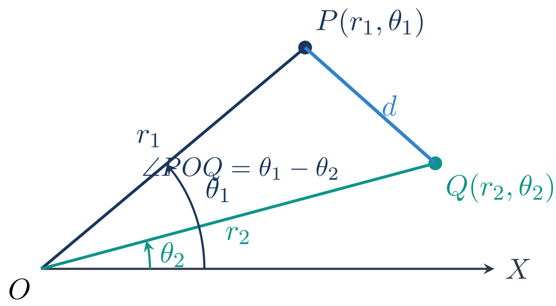
$$x^2 + y^2 = r^2 \cos^2\theta + r^2 \sin^2\theta = r^2 \implies r = \sqrt{x^2 + y^2} \tag{2}$$

Dividing y by x:

$$\tan\theta = \frac{PN}{ON} = \frac{y}{x} \implies \theta = \tan^{-1}\left(\frac{y}{x}\right) \tag{3}$$

2 Distance Between Two Points in Polar Coordinates

Let $P(r_1, \theta_1)$ and $Q(r_2, \theta_2)$ be two given points in polar coordinates with respect to the pole O and initial line OX.



In $\triangle POQ$:

- $OP = r_1, \quad OQ = r_2$
- $\angle POQ = |\theta_2 - \theta_1|$

By applying the **Law of Cosines** in $\triangle POQ$:

$$PQ^2 = OP^2 + OQ^2 - 2 \cdot OP \cdot OQ \cdot \cos(\angle POQ)$$

$$PQ^2 = r_1^2 + r_2^2 - 2r_1r_2\cos(\theta_2 - \theta_1)$$

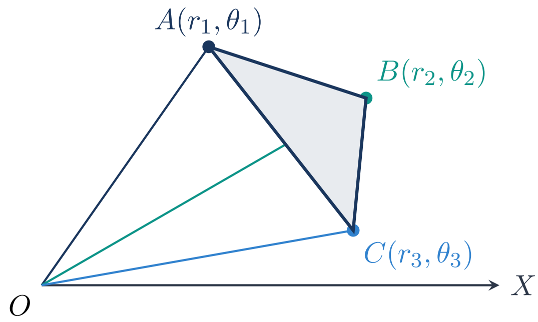
Polar Distance Formula

The distance $d = PQ$ between two points $P(r_1, \theta_1)$ and $Q(r_2, \theta_2)$ is given by:

$$PQ = \sqrt{r_1^2 + r_2^2 - 2r_1r_2\cos(\theta_2 - \theta_1)} \tag{4}$$

3 Area of a Triangle in Polar Coordinates

Consider a triangle $\triangle ABC$ with vertices in polar coordinates given by $A(r_1, \theta_1)$, $B(r_2, \theta_2)$, and $C(r_3, \theta_3)$.



The area of $\triangle ABC$ can be obtained by summing/subtracting the areas of triangles formed with the pole O:

$$\text{Area}(\triangle ABC) = \text{Area}(\triangle AOB) + \text{Area}(\triangle AOC) - \text{Area}(\triangle BOC)$$

Recall that the area of any triangle with sides a, b and included angle α is $\frac{1}{2}ab\sin\alpha$. Therefore:

$$\text{Area}(\triangle AOB) = \frac{1}{2}r_1 r_2 \sin(\theta_2 - \theta_1)$$

$$\text{Area}(\triangle BOC) = \frac{1}{2}r_2 r_3 \sin(\theta_3 - \theta_2)$$

$$\text{Area}(\triangle AOC) = \frac{1}{2}r_1 r_3 \sin(\theta_3 - \theta_1) = -\frac{1}{2}r_1 r_3 \sin(\theta_1 - \theta_3)$$

Combining these yields the general area formula:

Area of Triangle in Polar Coordinates

The area of triangle $\triangle ABC$ with vertices $A(r_1, \theta_1), B(r_2, \theta_2), C(r_3, \theta_3)$ is:

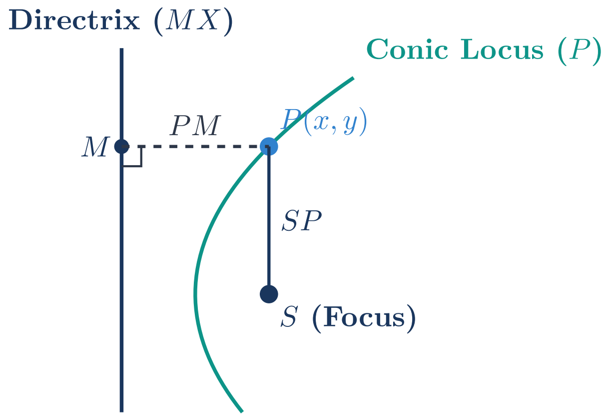
$$\text{Area}(\triangle ABC) = \frac{1}{2}\Big| r_1 r_2 \sin(\theta_2 - \theta_1) + r_2 r_3 \sin(\theta_3 - \theta_2) + r_3 r_1 \sin(\theta_1 - \theta_3) \Big| \tag{5}$$

4 Polar Equation of a Conic Section

Definition: Conic Section

A **conic section** (or simply conic) is defined as the locus of a moving point P in a plane such that the ratio of its distance from a fixed point S (called the **focus**) to its perpendicular distance from a fixed straight line MX (called the **directrix**) is always constant.
This constant ratio is called the **eccentricity** of the conic, denoted by e:

$$\frac{SP}{PM} = e \tag{6}$$



4.1 Classification of Conics based on Eccentricity (e)

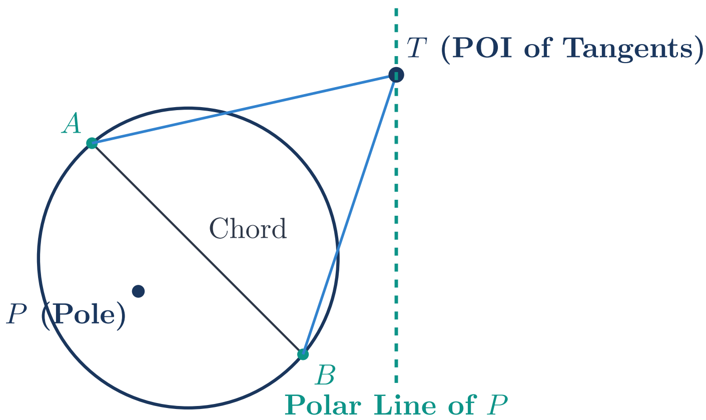
Depending on the value of eccentricity e, the locus of point P forms three fundamental conics:

1. **Parabola $(e = 1)$:** Here $SP = PM$. The locus is a parabola.
2. **Ellipse $(e < 1)$:** Here $SP < PM$. The locus is an ellipse.
3. **Hyperbola $(e > 1)$:** Here $SP > PM$. The locus is a hyperbola.

5 Theory of Pole and Polar

Definition: Pole and Polar

If chords of a conic section are drawn through a fixed point P, and tangents are drawn at the extremities (A and B) of all such chords, then the locus of the point of intersection of these two tangents is a straight line called the **Polar** of the fixed point P.
The fixed point P itself is called the **Pole** of that straight line.



6 Summary of Key Formulas

Analytic Geometry Quick Reference Sheet

1. **Cartesian to Polar:** $x = r\cos\theta, \quad y = r\sin\theta$
2. **Polar to Cartesian:** $r = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$
3. **Polar Distance Formula:** $d = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_2 - \theta_1)}$
4. **Area of Polar Triangle:**

$$\text{Area} = \frac{1}{2}\Big|r_1 r_2 \sin(\theta_2 - \theta_1) + r_2 r_3 \sin(\theta_3 - \theta_2) + r_3 r_1 \sin(\theta_1 - \theta_3)\Big|$$

5. **Conic Eccentricity Ratio:** $\frac{SP}{PM} = e \quad (\text{Parabola: } e = 1, \quad \text{Ellipse: } e < 1, \quad \text{Hyperbola: } e > 1)$